CHAPTER # 7

DISCRETE FOURIER TRANSFORM (DFT)

(if $X(\omega)$ is discrete - DFT)
(if $X(\omega)$ is continuous - DTFT)

Frequency analysis of a discrete time signals is usually performed on a digital signal processing. To perform frequency analysis on a time signal $\{x(n)\}$, we convert the time domain sequence to an equivalent frequency domain representation. We know that such a representation is given by the fourier transform $X(\omega)$ of the sequence $x(n)$. However, $X(\omega)$ is continuous function of frequency and therefore it is not a computationally ~~eq~~ convienient representation of the sequence $\{x(n)\}$, and therefore, we now consider the representation of a sequence $x(n)$ by samples of its spectrum $X(\omega)$. Such representation is called discrete fourier transform (DFT).

7.1 FREQUENCY-DOMAIN SAMPLING: The discrete fourier transform

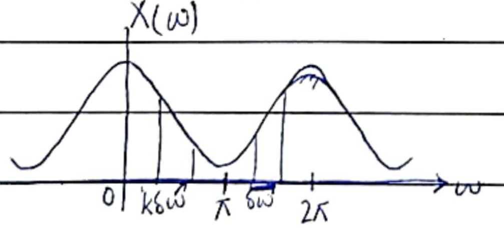
7.1.1 FREQUENCY DOMAIN SAMPLING OF DISCRETE-TIME SIGNALS:

Recall that

sampling frequency

I/p no. of samples

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \qquad (7.1.1)$$



$\delta\omega$ - distance b/w two samples.

Suppose that we sample $X(\omega)$ periodically in frequency at spacing of $\delta\omega$ radians between successive samples, where $X(\omega)$ is periodic with period 2π.

⇒ For convenience, we take N equidistant samples in the interval $0 \le \omega < 2\pi$ (after 2π, it repeats) with spacing $\delta\omega = 2\pi/N$, as shown in figure.

⇒ If we evaluate (7.1.1) at $\underbrace{\omega = 2\pi k/N}_{\text{equidistant samples}}$ we obtain,

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\frac{2\pi k}{N}} \quad ; k = 0, 1, \dots N-1$$

13/3 DFT: $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi k}{N}}$ (i) ; $k = 0, 1, 2, \dots N-1$

IDFT: $x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi k}{N}}$ (ii) ; $n = 0, 1, 2, \dots N-1$

THE DFT AS A LINEAR TRANSFORMATION:

The formulas in eq (i) and eq (ii) may be expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad ; k = 0, 1, 2, \dots N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k) W_N^{-kn} \quad ; k = 0, 1, 2, \dots N-1.$$

where, $W_N = e^{\frac{j2\pi}{N}}$

~~calculate~~

Q. 4-POINT DFT:

$N = 4$, $k = n = 0:3$

⇒ For

$k = 0$

⇒ $\underbrace{x(0) W_4^0 + x(1) W_4^0 + x(2) W_4^0 + x(3) W_4^0}_{x_N \Rightarrow 4\times1}$,

⇒ $k = 1$

⇒ $x(0) W_4^0 + x(1) W_4^1 + x(2) W_4^2 + x(3) W_4^3$

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$k=2$ $\quad kn=2(1) \quad kn=2(2)$

$\Rightarrow x(0)W_4^0 + x(1)W_4^2 + x(2)W_4^4 + x(3)W_4^6$

$k=3$

$\Rightarrow x(0)W_4^0 + x(1)W_4^3 + x(2)W_4^6 + x(3)W_4^9$

$\underset{4\times4}{W_N}\ \underset{4\times1}{x_N} = X(k)_{4\times1}$ → 4 point DFT

$$W_N = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}_{4\times4} \qquad \text{(iii)}$$

($\because 4\times4 \Rightarrow$ 4-Point)

(if 5-Point $\Rightarrow 5\times5$)

$W_4^0 = ?$

$W_4^1 = ?$

$W_4^2 = ?$

$W_4^3 = ?$

$W_4^0 = W_4^{kn} = e^{\frac{-j2\pi(0)}{4}} = 1$

$W_4^1 = W_4^{kn} = e^{\frac{-j2\pi(1)}{4}} = e^{-j\frac{\pi}{2}} = \cos(\pi/2) - j\sin(\pi/2)$

$= 0 - j(1) = -j$

$W_4^2 = W_4^{kn} = e^{\frac{-j2\pi(2)}{4}} = e^{-j\pi} = \cos(\pi) - j\sin(\pi)$

$= -1 - 0 = -1$

$W_4^3 = e^{\frac{-j2\pi(3)}{4}} = e^{-j\frac{3\pi}{2}} = \cos(3\pi/2) - j\sin(3\pi/2)$

$= 0 - j(-1) = j$

By periodicity property $\quad\quad \because$ Periodicity: $W_N^k = W_N^{k+N}$

$W_4^0 = W_4^4$

$W_4^1 = W_4^5 = W_4^9$ $\quad \because 4+1=5$, $\because 5+4=9$

$W_4^2 = W_4^6$

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if Compute the DFT of the 4-point sequence:
$x(n) = [0 \quad 1] \Rightarrow 1\times2 \Rightarrow [0 \quad 1 \quad 0 \quad 0]^T = 4\times4$
& $W_N \Rightarrow 4\times4$

* Periodic → sequence repeat
Aperiodic → do not repeat

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$\Rightarrow$ (iii)
$$W_N = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

EXAMPLE # 5.1.3
Compute the DFT of the 4-point sequence:
$$x(n) = [0 \quad 1 \quad 2 \quad 3]$$

$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}_{4\times4} \times \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}_{4\times1} = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix} = X(k)$$